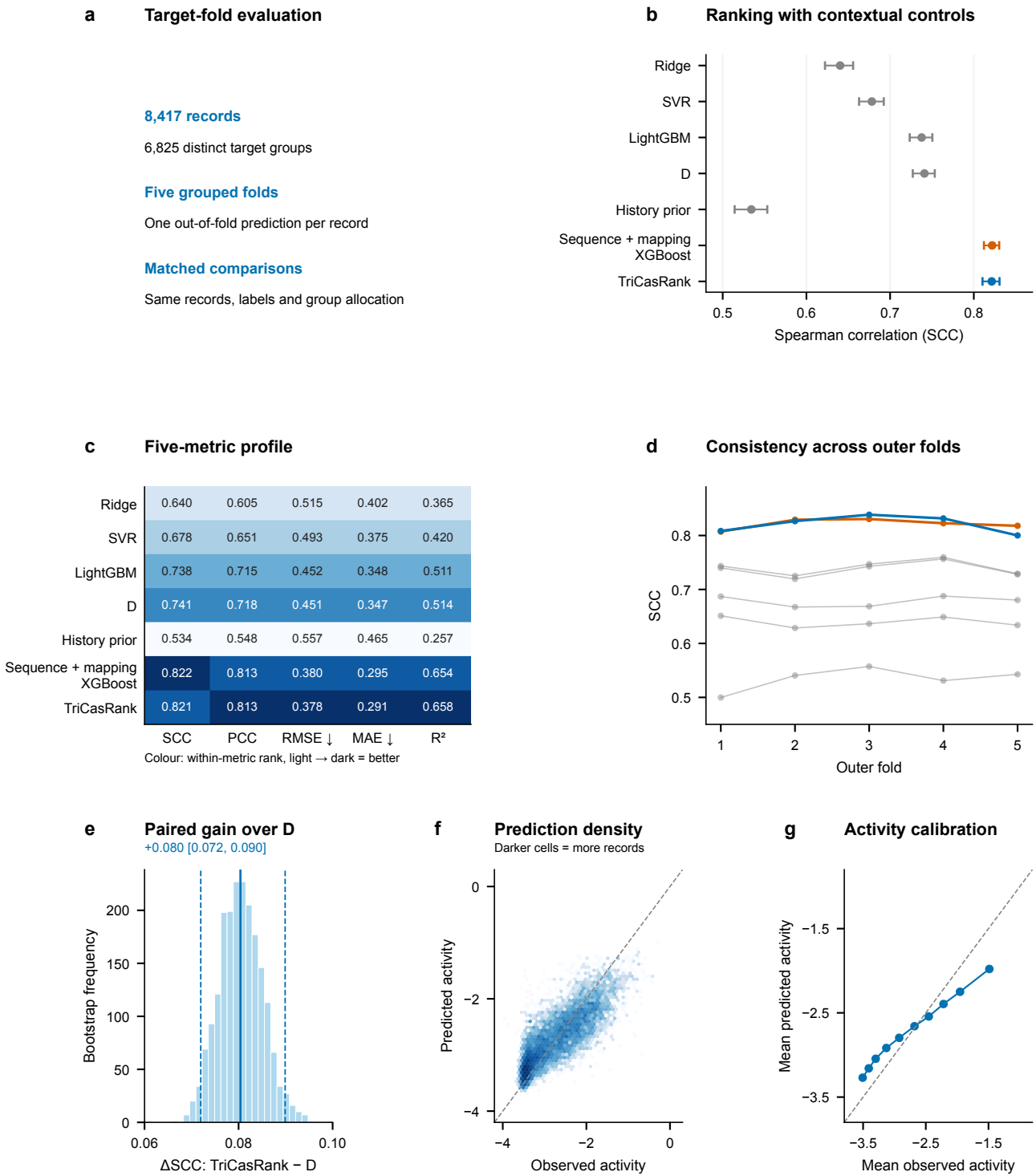

a
Target-fold evaluation
8,417 records
6,825 distinct target groups
Five grouped folds
One out-of-fold prediction per record
Matched comparisons
Same records, labels and group allocation
b
Ranking with contextual controls
Ridge
SVR
LightGBM
D
History prior
Sequence + mapping XGBoost
TriCasRank
0.5
0.6
0.7
0.8
Spearman correlation (SCC)
c
Five-metric profile
Ridge 0.640 0.605 0.515 0.402 0.365
SVR 0.678 0.651 0.493 0.375 0.420
LightGBM 0.738 0.715 0.452 0.348 0.511
D 0.741 0.718 0.451 0.347 0.514
History prior 0.534 0.548 0.557 0.465 0.257
Sequence + mapping XGBoost 0.822 0.813 0.380 0.295 0.654
TriCasRank 0.821 0.813 0.378 0.291 0.658
SCC PCC RMSE ↓ MAE ↓ R²
Colour: within-metric rank, light → dark = better
d
Consistency across outer folds
SCC
0.5
0.6
0.7
0.8
1
2
3
4
5
Outer fold
e
Paired gain over D
+0.080 [0.072, 0.090]
Bootstrap frequency
0
50
100
150
200
0.06
0.08
0.10
ΔSCC: TriCasRank − D
f
Prediction density
Darker cells = more records
Predicted activity
0
−2
−4
−4
−2
0
Observed activity
g
Activity calibration
Mean predicted activity
−1.5
−2.5
−3.5
−3.5
−2.5
−1.5
Mean observed activity